Udbhav Saxena

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Summary

Machine Learning Engineer/ Applied AI Engineer with 3 years building production clinical AI spanning LLMs, RAG, full-stack ML systems, speech pipelines, and EHR integration. Shipped evaluated, production-grade systems reducing clinical documentation time by 70% across behavioral health workflows.

Work Experience

Machine Learning Engineer/ Applied AI Engineer, Civitas Health Services **2023 – Present**

- **Agentic Clinical Documentation:** Built and shipped an agentic LLM system (OpenAI function calling, custom orchestration) for autonomous generation of progress notes, ISP summaries, and SMART goals; reduced average note completion time by ∼70%.
- **Full-Stack EHR Platform:** Architected and shipped a clinical EHR platform (React, PHP, EspoCRM) supporting 7 workflow stages; integrated 3 external APIs (OpenAI, Twilio, Phaxio) and built custom React portals, drag-and-drop planners, and assessment tools for behavioral health staff.
- **Survey-to-Assessment NLP Pipeline:** Designed a pipeline (spaCy, BERT) converting 4 instruments (CSSRS, SBIRT, FAST, VBMAPP) into structured draft assessments; eliminated ∼30 min of manual write-up per intake.
- **Call Transcription & PHI Redaction:** Built a transcription pipeline (Twilio + OpenAI Whisper) with entity extraction, PHI redaction, and risk-phrase detection; processes 100+ calls/month into structured EHR entries.
- **RAG for Medicaid Guidelines:** Built a retrieval-augmented generation layer (pgvector, OpenAI embeddings, Flask) for inline guideline lookup; cut authorization prep time by ∼40%; enabled conversational patient-document querying during ISP authoring.

AI Engineer, Intern, Volvo Group North America **2022**

- **Packaging Optimization:** Built XGBoost model (85% accuracy) deployed across 2 production lines; generated $350K in annualized cost savings (6:1 ROI); parts-redundancy recommender reduced master database by 20%.
- **Time-Series & Computer Vision:** Built ARIMA/SARIMA time-series forecasting (±5-unit error) and YOLOv4 tool detection model for cobot automation; reduced false-positive safety stops and improved line throughput.

Projects

- **Clinical RAG Pipeline:** End-to-end retrieval-augmented generation system over behavioral health guidelines; chunking, embedding (OpenAI), pgvector storage, and re-ranking; grounded LLM outputs in source documents with citation traces.
- **BERT Fine-tune & Compression:** 94% accuracy on text classification; applied DistilBERT distillation, INT8 quantization, and ONNX Runtime optimization; reduced inference latency by ∼3× with minimal accuracy loss.
- **CRNN for OCR:** 95% test accuracy on sequence recognition; full preprocessing and augmentation pipeline with CTC loss decoding.

Skills

- **ML / NLP:** PyTorch, TensorFlow, Hugging Face (BERT, GPT-2, DistilBERT), spaCy, NLTK, ONNX Runtime, Gensim, XGBoost, ARIMA/SARIMA
- **LLM & Agents:** Agentic workflows, RAG, model evaluation, structured outputs, fine-tuning, prompt engineering, LLM chaining, OpenAI & Anthropic APIs
- **Speech / ASR:** OpenAI Whisper, Twilio, PHI redaction, call transcription & summarization
- **Infrastructure:** Docker, Kubernetes, Flask, pgvector, PySpark, Pandas, NumPy, Git, Linux
- **Full-Stack ML:** React, PHP, EspoCRM, REST APIs, Stripe, Phaxio, end-to-end ML system design
- **Languages:** Python, C/C++, R, MATLAB

Education

M.S. Engineering Science, University at Buffalo, SUNY **2021 – 2023**
B.Tech. Mechanical Engineering, Shiv Nadar University **2017 – 2021**